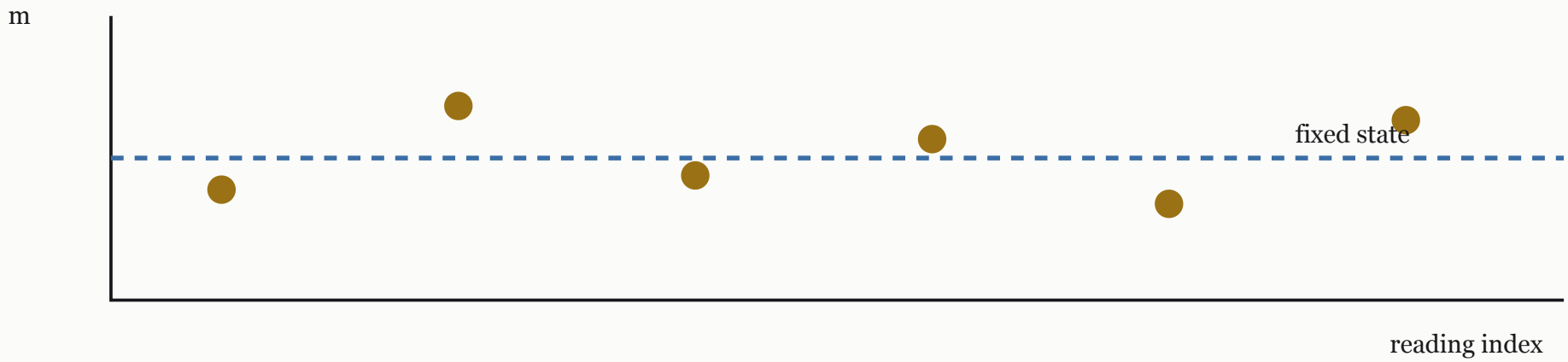
WHY

Uncertainty

Which position should a noisy sensor report?



> One stationary object. Several different readings.

`state` · `measurements` · `inferred answer`

Report an estimate and the uncertainty around it.

WHY

Uncertainty

What does a distribution add to a model?

$$z = x + \varepsilon$$

- x unknown state (m)
- z observed reading (m)
- ε random error (m)

Deterministic prediction

One output for specified inputs.

Probabilistic description

Possible outcomes and their weights.

> Uncertainty describes the model, not a moving wall.

Random variables

How does an outcome become a number?

$$x : \Omega \to \mathbb{R}, \qquad x(\omega) \in \{-1, 0, 2\}$$

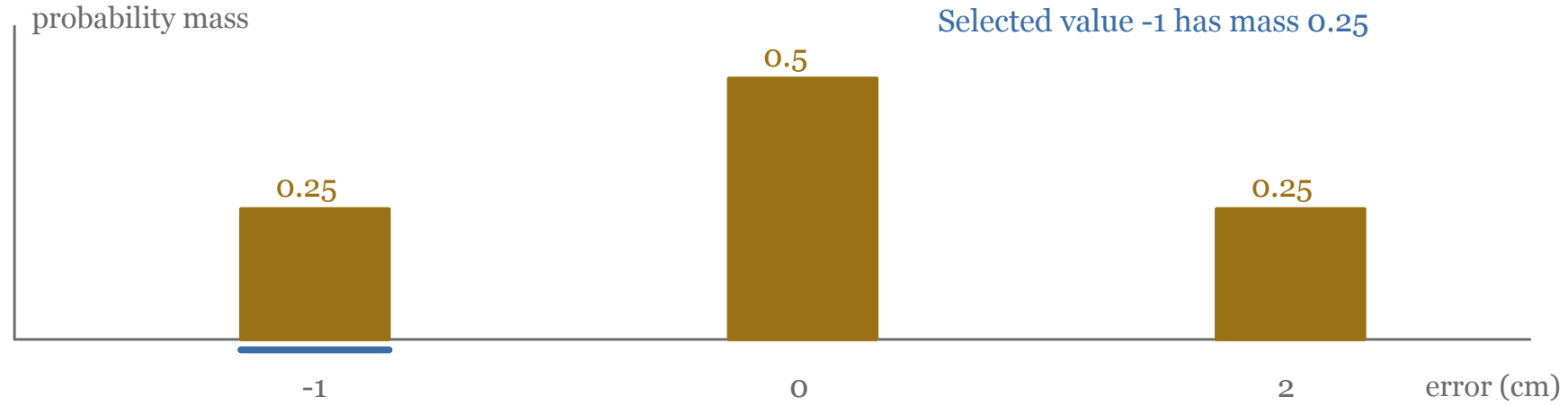
- Ω possible outcomes
- ω one outcome
- x random variable: numerical error (cm)

Outcome A → −1 cm	Outcome B → 0 cm	Outcome C → 2 cm

> The rule is fixed; the outcome is uncertain.

PMF and PDF

How do probabilities attach to discrete values?



$$p(a) = P(x = a), \qquad p(a) \geq 0, \qquad \sum_{a} p(a) = 1$$

- x discrete random error
- a possible value (cm)
- $p(a)$ probability mass
- P event probability

DEFINITION AND REASONING

PMF and PDF

What does the three-point error model predict?

WORKED EXAMPLE E1

Error values −1, 0, 2 cm; masses ¼, ½, ¼. Find nonnegative-error probability and mean.

$$P(x \geq 0) = \tfrac{1}{2} + \tfrac{1}{4} = \tfrac{3}{4}, \qquad \mu = (-1)\tfrac{1}{4} + 0\tfrac{1}{2} + 2\tfrac{1}{4} = \tfrac{1}{4}$$

- P event probability
- x error (cm)
- μ mean error (cm)

TRY, COMPARE, EXPLAIN

PMF and PDF

Can you recompute an event after weights change?

Same errors -1, 0, 2 cm; new masses $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{4}$.

Check normalization. Find the probability of nonnegative error.

PMF and PDF

Why does a continuous event need an area?

$$P(a \le x \le b) = \int_a^b p(t)\,dt$$

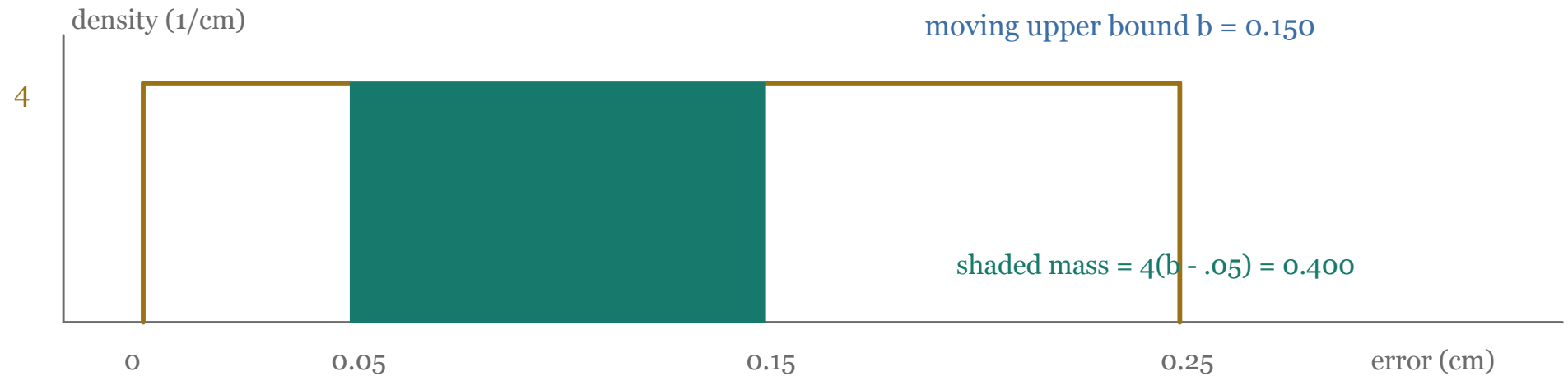
- x continuous random quantity
- a, b interval endpoints
- t integration variable
- $p(t)$ density, inverse units of x

$$P(x = c) = \int_c^c p(t)\,dt = 0$$

- c one fixed value
- P probability for a variable with a density

PMF and PDF

Can a valid density be greater than one?



WORKED EXAMPLE E2 · UNIFORM ERROR

$$p(t) = 4 \;\; (0 \le t \le .25), \qquad P(.05 \le x \le .15) = 4(.15 - .05) = .4$$

- t, x error (cm)
- p density (cm^{-1})
- P interval probability

TRY, COMPARE, EXPLAIN

PMF and PDF

What changes when the uniform interval doubles?

Uniform error on [0, .5] cm. Find the density and $P(.1 \le x \le .3)$.

Moments

How do we average a function of a random value?

$$\mathbb{E}[g(x)] = \sum_{a} g(a)p(a)$$

- x discrete random variable
- a support value
- g function being averaged
- p PMF

$$\mathbb{E}[g(x)] = \int_{-\infty}^{\infty} g(t)p(t)\,dt$$

- x continuous random variable
- t integration variable
- p density
- g assume E[|g(x)|] is finite

TRY, COMPARE, EXPLAIN

Moments

Variance or second moment?

$$\mathrm{Var}(x) = \mathbb{E}[(x-\mu)^2] = \mathbb{E}[x^2] - \mu^2$$

x	random error (cm)
$\mu = \mathbb{E}[x]$	mean (cm)
$\mathrm{Var}(x)$	variance (cm²)

E3 / T3

Values −1,0,2 cm. E3: ¼,½,¼; T3: ½,¼,¼. Find μ and variance.

DEFINITION AND REASONING

Moments

What is the mean of a random column vector?

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \qquad \mu = \mathbb{E}[x] = \begin{bmatrix} \mathbb{E}[x_1] \\ \mathbb{E}[x_2] \end{bmatrix}$$

- x random column in $\mathbb{R}^2$
- x_i component i
- μ mean column in $\mathbb{R}^2$

> Average corresponding components; preserve the column shape.

Samples use superscripts: $x^{(s)}$. Components use subscripts: x_i.

DEFINITION AND REASONING

Moments

Why does covariance use an outer product?

$$\Sigma = \mathbb{E}[(x-\mu)(x-\mu)^T], \qquad \Sigma_{ij} = \mathbb{E}[(x_i-\mu_i)(x_j-\mu_j)]$$

- x, μ — random column and mean in R^d
- Σ — population covariance, d × d
- i, j — coordinate indices

> Diagonal: component variance. Off-diagonal: paired deviations.

DEFINITION AND REASONING

Moments

Why is covariance positive semidefinite?

$$a^T \Sigma a = \mathbb{E}[(a^T(x-\mu))^2] \geq 0$$

- a fixed direction column in $\mathbb{R}^d$
- x, μ random column and population mean
- Σ population covariance

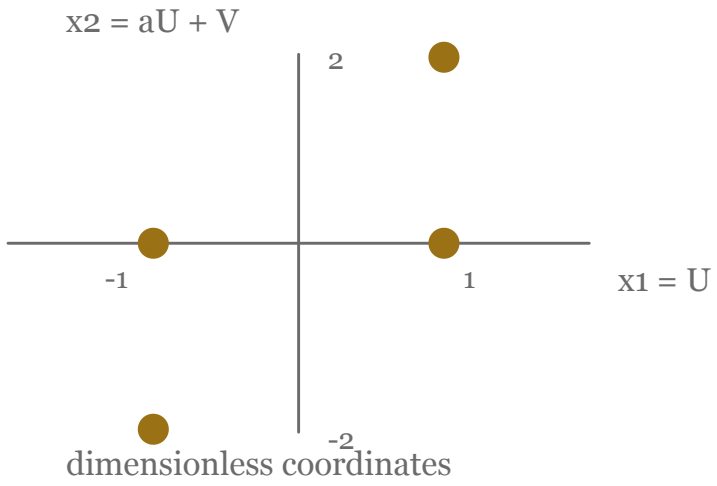
$$C = \frac{X_c^T X_c}{N-1} \quad (N > 1)$$

- C sample covariance estimate
- X_c N centered sample rows
- N number of samples

DEFINITION AND REASONING

Moments

How do independent signs create correlation?



a = 1.00

four equally likely outcomes

a: 0 to 1 adds the shared sign

WORKED EXAMPLE E4

Independent fair signs U,V. Let $x = (U, U+V)^T$.

$$\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \qquad \Sigma = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$

U, V independent, equally likely ± 1

x, μ random column and mean

Σ population covariance

TRY, COMPARE, EXPLAIN

Moments

What changes when the shared sign is doubled?

Independent fair signs U,V; $x = (U, 2U + V)^T$. Find the mean and covariance.

DEFINITION AND REASONING

Moments

Does zero covariance guarantee independence?

WORKED EXAMPLE E5

x is uniform on {−1,0,1}; y=x². Is y independent of x?

$$\operatorname{Cov}(x, y) = 0, \qquad P(y = 0 \mid x = 0) = 1 \neq \tfrac{1}{3} = P(y = 0)$$

x, y	random scalars
P	event probability
Cov	population covariance

Gaussian geometry

What defines a multivariate Gaussian density?

$$p(x) = (2\pi)^{-d/2} |\Sigma|^{-1/2} \exp(-d_M^2/2)$$

- x, μ random column and mean in R^{d}
- d dimension
- $\Sigma \succ 0$ positive definite covariance
- $|\Sigma|$ determinant
- p density
- d_M^2 squared Mahalanobis distance

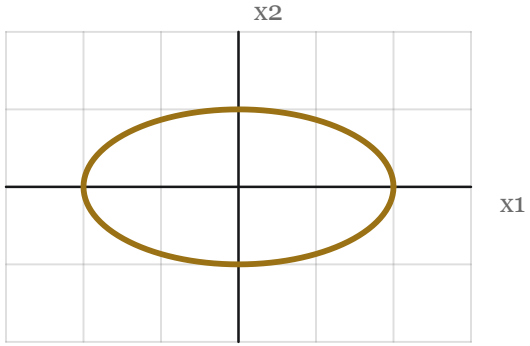
$$d_M^2 = (x-\mu)^T \Sigma^{-1} (x-\mu)$$

- d_M Mahalanobis distance
- Σ^{-1} inverse covariance
- $x - \mu$ centered displacement

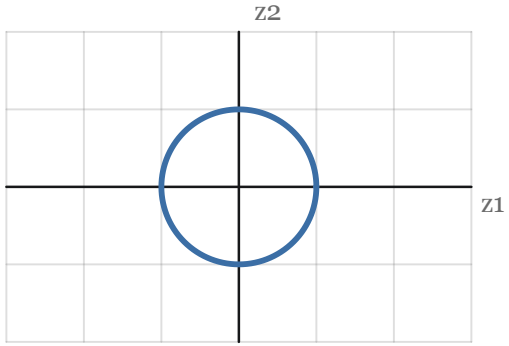
Gaussian geometry

Rotate, then standardize each coordinate

centered original coordinates



divide x1 by its standard deviation

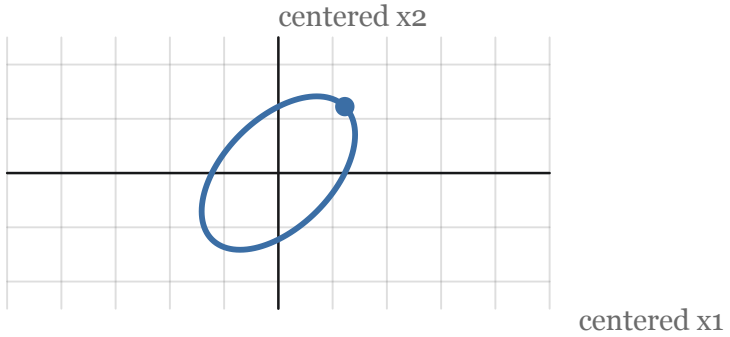


$$\Sigma = Q\Lambda Q^T, \quad z = \Lambda^{-1/2} Q^T (x - \mu)$$

Symbols Q eigenvectors; Λ positive eigenvalues; z whitened displacement; x, μ position and mean.

Gaussian geometry

A contour radius becomes an ellipse axis



unit Mahalanobis contour

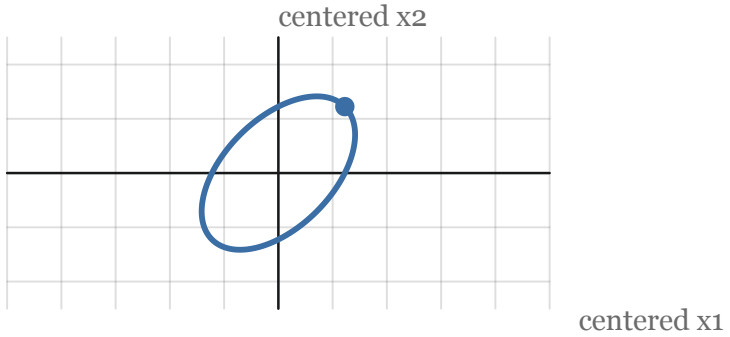
eigenvalues 3 and 1

$$x = \mu + Q\Lambda^{1/2} z, \qquad \|z\| = c$$

Symbols c standardized radius; $\sqrt{\lambda_i}$ standard deviation along eigenvector q_i.

Gaussian geometry

Equal lengths can have unequal surprise



unit Mahalanobis contour

eigenvalues 3 and 1

$$\Sigma = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad \mu = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Symbols Σ covariance; μ mean; compare $x = (2, 0)^T$ and $y = (2, -2)^T$.

WORKED EXAMPLE

$$d_M^2(x) = \tfrac{2}{3}, \qquad d_M^2(y) = 2$$

Result d_M covariance-scaled distance.

TRY, COMPARE, EXPLAIN

Gaussian geometry

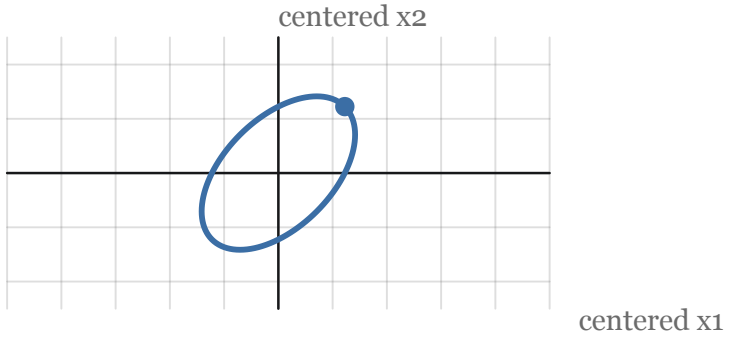
Try two directions with one covariance

$$\mu = 0, \quad \Sigma = \operatorname{diag}(4, 1), \quad u = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \; v = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$$

Symbols Find both d_M^2 values and the density ratio $p(u)/p(v)$.

Gaussian geometry

A unit ellipse contains less than half



unit Mahalanobis contour

eigenvalues 3 and 1

$$P(d_M^2 \le c^2) = 1 - e^{-c^2/2} \quad (d = 2)$$

Symbols c contour radius; $d = 2$ dimension; the formula assumes a nonsingular Gaussian.

WORKED EXAMPLE

$$c_{.95} = \sqrt{-2 \log .05} \approx 2.44775$$

Result $c_{.95}$ radius enclosing 95%.

Bayes

Which population does a positive select?

all model trials: 10,000



event 100 non-event 9,900

Conditioning selects a new denominator

$$P(H \mid +) \text{ need not equal } P(+ \mid H)$$

Symbols H event of interest; $+$ positive sensor result; the bar restricts the population.

Bayes

Write the joint event in two ways

$$P(H \mid +) = \frac{P(+ \mid H)P(H)}{P(+)}$$

Symbols $P(H)$ prior; $P(+ \mid H)$ likelihood; $P(+) > 0$ evidence.

Bayes

Normalize over both sources of positives

$$P(+) = P(+ \mid H)P(H) + P(+ \mid \neg H)P(\neg H)$$

Symbols $\neg H$ complement; the two joint events are disjoint and exhaustive.

Bayes

Rare events can lose the positive majority

all model trials: 10,000



event 100 non-event 9,900

Conditioning selects a new denominator

$$P(H) = .01, \quad P(+ \mid H) = .9, \quad P(+ \mid \neg H) = .1$$

Symbols 10,000 model trials: 100 event cases and 9,900 non-event cases.

WORKED EXAMPLE

$$P(H \mid +) = \frac{90}{90 + 990} = \frac{1}{12}$$

Result H event; $+$ positive reading.

TRY, COMPARE, EXPLAIN

Bayes

Change the base rate, keep the sensor

$$P(H) = .1, \quad P(+ \mid H) = .9, \quad P(+ \mid \neg H) = .1$$

Symbols Use 10,000 model trials again. Find $P(H \mid +)$.

Break

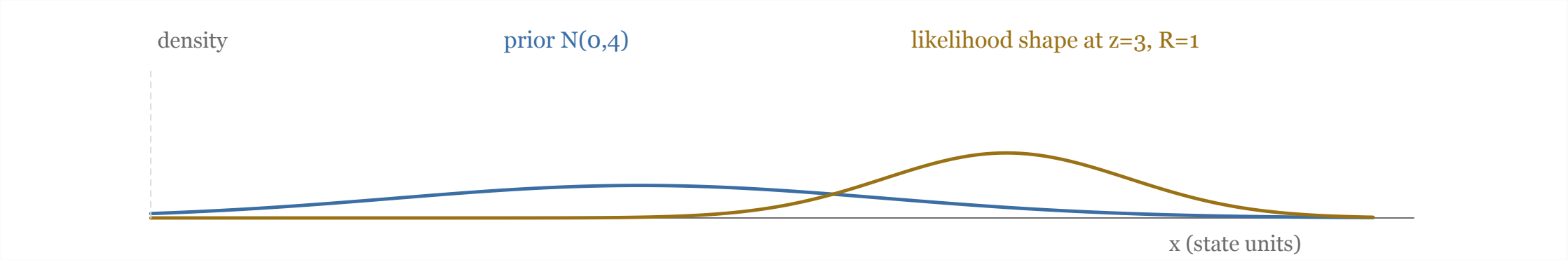
Pause: three different kinds of probability

$$P(X \in A), \qquad p(x), \qquad P(H \mid +)$$

Symbols A event; p density; conditioning changes the reference population.

Gaussian conditioning

A numeric reading updates a curve



$$x \sim \mathcal{N}(\mu_0, P_0), \quad z = x + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, R)$$

Symbols x uncertain state; z observed value; $P_0, R > 0$ variances; independent noise.

Gaussian conditioning

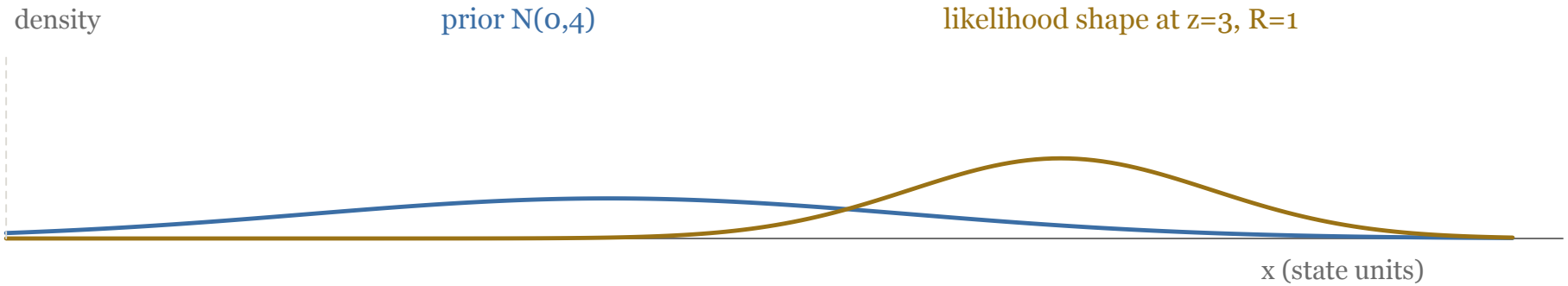
Add precisions, then complete the square

$$P_1^{-1} = P_0^{-1} + R^{-1}, \quad \mu_1 = P_1(P_0^{-1}\mu_0 + R^{-1}z)$$

Symbols P_1, μ_1 posterior variance and mean; inverse variance is precision.

Gaussian conditioning

The gain weights the innovation

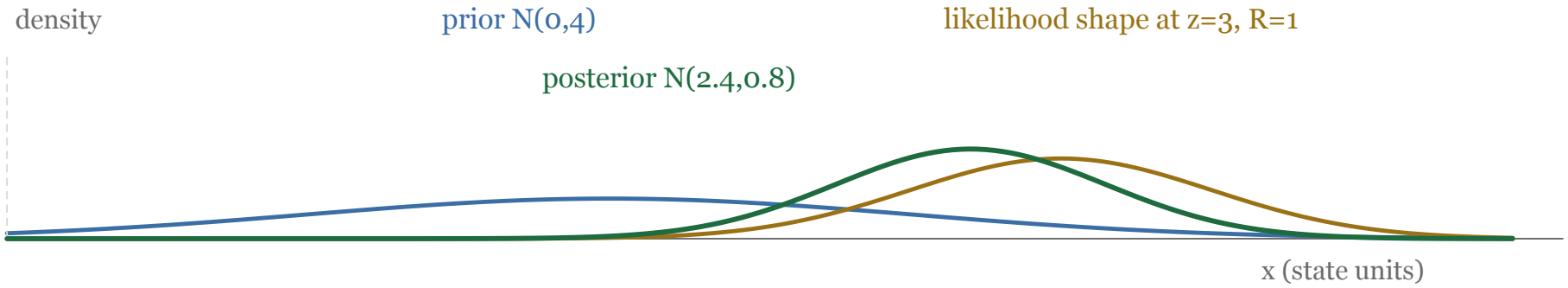


$$K = \frac{P_0}{P_0 + R}, \quad \mu_1 = \mu_0 + K(z - \mu_0), \quad P_1 = (1 - K)P_0$$

Symbols K gain; $z - \mu_0$ innovation; $0 < K < 1$ for positive variances.

Gaussian conditioning

Fuse a prior and a precise reading



$$\mu_0 = 0, \quad P_0 = 4, \quad z = 3, \quad R = 1$$

Symbols P_0, R are variances, not standard deviations.

WORKED EXAMPLE

$$K = \tfrac{4}{5}, \qquad \mu_1 = 2.4, \qquad P_1 = .8$$

Result K gain; μ_1, P_1 posterior mean, variance.

TRY, COMPARE, EXPLAIN

Gaussian conditioning

What if the same reading is noisier?

$$\mu_0 = 0, \quad P_0 = 4, \quad z = 3, \quad R = 4$$

Symbols Predict the direction of change, then compute K, μ_1, P_1.

Gaussian conditioning

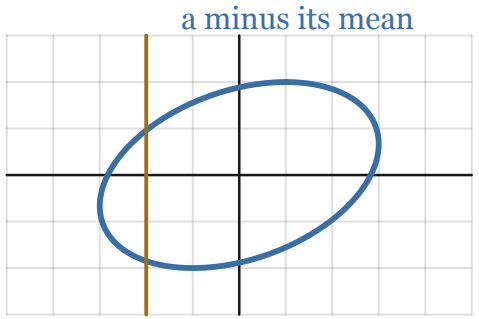
A correlated block predicts another block

$$\mu_{a|b} = \mu_a + \Sigma_{ab}\Sigma_{bb}^{-1}(b - \mu_b)$$

Symbols Jointly Gaussian blocks; invertible Σ_{bb}. $a \in \mathbb{R}^p$, $b \in \mathbb{R}^q$; $\Sigma_{ab} : p \times q$.

Gaussian conditioning

Conditioning removes predictable variance



slice fixes b; a remains uncertain

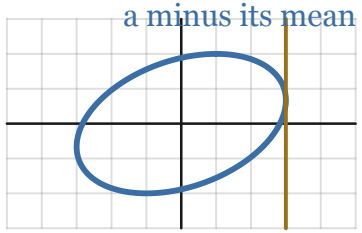
$$\Sigma_{a|b} = \Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}$$

Symbols Σ_{aa} marginal covariance; $\Sigma_{ba} = \Sigma_{ab}^{T}$; joint Gaussian, invertible Σ_{bb}.

TRY, COMPARE, EXPLAIN

Gaussian conditioning

Slice the same Gaussian at two readings



slice fixes b; a remains uncertain

$$\mu = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \Sigma = \begin{bmatrix} 4 & 2 \\ 2 & 9 \end{bmatrix}, \quad b = 5$$

Symbols a, b scalar blocks. For $b = -1$, find the conditional mean and variance.

Estimation

Choose a parameter to explain fixed data

$$b = A\theta + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2 I)$$

Symbols $b \in \mathbb{R}^N$ data; $A \in \mathbb{R}^{N \times p}$ known; $\theta \in \mathbb{R}^p$ unknown; σ^2 known.

Estimation

Gaussian products become a squared loss

$$-\log p(b \mid \theta) = \frac{N}{2}\log(2\pi\sigma^2) + \frac{\|b - A\theta\|^2}{2\sigma^2}$$

Symbols N number of readings; σ^2 common variance; residual $r = b - A\theta$.

Estimation

The familiar normal equation returns

$$A^T A\hat{\theta} = A^T b, \qquad A^T(b - A\hat{\theta}) = 0$$

Symbols $\hat{\theta}$ MLE; full column rank gives uniqueness; residual is orthogonal to columns.

Estimation

Fit three readings with two coefficients

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

Symbols $\theta = (\text{intercept}, \text{slope})^T$; $r = b - A\hat{\theta}$.

WORKED EXAMPLE

$$\hat{\theta} = \begin{pmatrix} 7/6 \\ 1/2 \end{pmatrix}, \quad r = \begin{bmatrix} -1/6 \\ 1/3 \\ -1/6 \end{bmatrix}$$

Result $\hat{\theta}$ estimate; r observed minus fitted.

Estimation

A Gaussian prior sets a penalty scale

$$J(\theta) = \tfrac{1}{2}\|b - A\theta\|^2 + \tfrac{\lambda^2}{2}\|\theta - \mu_0\|^2, \quad \lambda^2 = \sigma^2/\tau^2$$

Symbols $\theta \sim \mathcal{N}(\mu_0, \tau^2 I)$ prior; σ^2 noise variance; λ^2 relative strength.

Estimation

The prior changes the normal equation

$$(A^T A + \lambda^2 I)\hat{\theta} = A^T b + \lambda^2 \mu_0$$

Symbols $\lambda^2 > 0$ gives a unique MAP; use the previous fit with $\mu_0 = 0, \lambda^2 = 1$.

WORKED EXAMPLE

$$\hat{\theta}_{MAP} = \begin{pmatrix} 4/5 \\ 3/5 \end{pmatrix}, \quad \text{SSE} = \tfrac{2}{5}$$

Result $\hat{\theta}_{MAP}$ posterior mode; SSE squared residual.

TRY, COMPARE, EXPLAIN

Estimation

A proper prior resolves an ambiguous fit

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}, \quad b = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \mu_0 = 0,\ \lambda^2 = 1$$

Task Find all least-squares fits, then the unique MAP estimate.

Integrated board

Calibrate one offset from three readings

$$b = (1, 2, 3)^T, \quad b_i = \theta + \varepsilon_i, \quad \varepsilon_i \overset{iid}{\sim} \mathcal{N}(0, 1)$$

Symbols θ offset; prior $\mathcal{N}(0, 1)$. Find MLE, posterior, CI and credible interval.

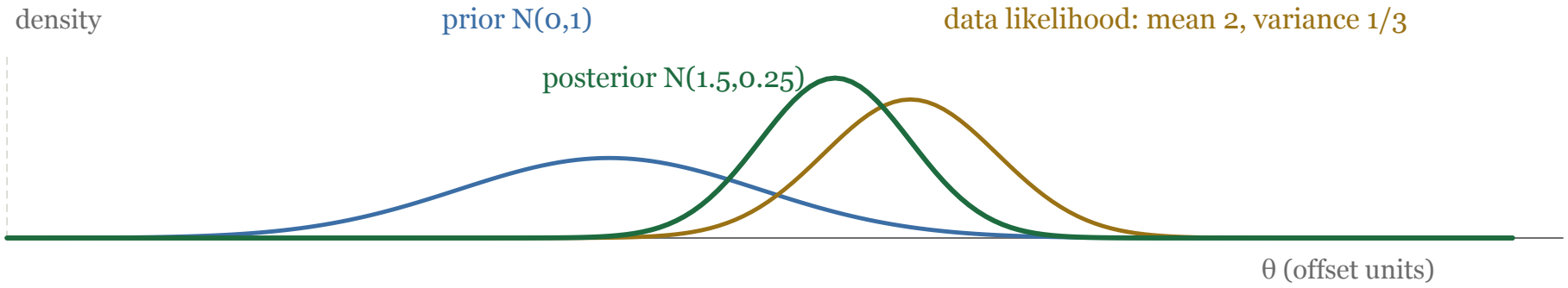
WORKED EXAMPLE

$$J_{data} = \tfrac{1}{2}(3\theta^2 - 12\theta + 14), \quad \hat{\theta}_{MLE} = 2$$

Result J_{data} squared-error objective; MLE its minimizer.

Integrated board

Complete the posterior square



$$J_{post} = \tfrac{1}{2}(4\theta^2 - 12\theta + 14) = 2(\theta - 1.5)^2 + 2.5$$

Symbols J_{post} negative log posterior up to a constant; Gaussian exponent uses $1/(2P)$.

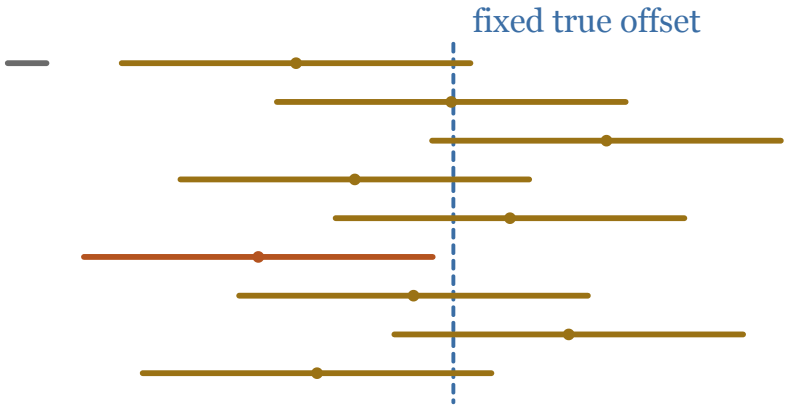
WORKED EXAMPLE

$$\theta \mid b \sim \mathcal{N}(1.5, .25), \qquad \theta_{MAP} = 1.5$$

Result θ_{MAP} posterior mode.

Intervals

Confidence belongs to a repeated procedure



illustrative repeated intervals (not a coverage estimate)

$$P_\theta(\theta \in C(B)) = .95, \quad C(B) = \bar{B} \pm 1.95996\,\sigma/\sqrt{N}$$

Symbols θ fixed; B repeated random data; $C(B)$ random interval; normal independent noise, known σ.

Intervals

Credibility belongs to the posterior

$$P(\theta \in I(b) \mid b) = .95$$

Symbols b observed and fixed; $\theta \mid b$ posterior random variable; prior and likelihood specified.

Intervals

The same readings yield different intervals

$$C(b) = 2 \pm 1.95996/\sqrt{3}, \quad I(b) = 1.5 \pm 1.95996(.5)$$

Symbols C known-noise confidence interval; I equal-tail posterior credible interval.

WORKED EXAMPLE

$$C = [.8684, 3.1316], \qquad I = [.5200, 2.4800]$$

Result C confidence; I credible.

TRY, COMPARE, EXPLAIN

Transfer

Calibrate new data independently

$$b = (2, 3)^T, \quad \sigma^2 = 1, \quad \theta \sim \mathcal{N}(1, 1)$$

Task Find MLE, posterior and both 95% intervals. Interpret each.

Intervals

What does a sensor specification promise?

$$\text{“}\pm 2\ \text{cm},\ 95\%\text{”}$$

Symbols Coverage? Posterior mass? A future reading? Manufacturing tolerance?

Synthesis

Explain the bridge without a formula sheet

$$\operatorname{Var}(X) = E[X^2] - E[X]^2, \qquad p(\theta \mid b) \propto p(b \mid \theta)p(\theta)$$

Symbols Connect moments → geometry → evidence → least squares → intervals.

References

Sources and optional extensions

1. Textbook · Goodfellow et al., ch.3
2. Textbook · Goodfellow et al., §5.5–5.6
3. Course notes · UBC Gaussian, pp.5,7–8
4. NIST · Confidence intervals
5. Optional course · MIT probability